

Pollution mapping for LSAP and dioceses

Analysis specification and map production workflows

6 October 2026

Build five coordinated map views: annual PM_{2.5}, annual NO₂, freshwater quality summarized by diocese, river plastic emissions related to LSAP member dots, and landfills and dumpsites summarized by diocesan count and area with member proximity. Preserve ecclesiastical context in every map and use copies of all foundational LSAP and ecclesiastical inputs.

Shared data architecture

Create a dedicated Pollution project folder containing FoundationCopies.gdb, PollutionAnalysis.gdb, restricted FreshwaterScratch.gdb, map styles, source metadata and final documentation. Copy Features or Export Features into FoundationCopies.gdb before analysis. Never repair geometry, calculate fields, join, or overwrite the source datasets in Diodata or LSAP Data.

Use the approved diocesan boundaries and member point layer from the existing atlas projects. Confirm their exact feature classes, source dates and coordinate systems before copying. Preserve stable diocese and member keys, ecclesiastical names and approved context layers. Retain member geocoding precision: city and country centroids cannot support precise neighbourhood proximity claims.

Validate copied geometry and boundary overlap. Resolve duplicate identifiers and document rules for overlapping jurisdictions, enclaves, boundary points and unassigned records. Use equal-area measurement for footprints and area-weighted statistics; use geodesic distance for global proximity. Preserve an all-dioceses output with nulls where evidence is missing.

Pollution palette

Colour	Hex	Map role
Pale stone	#E1E2DB	Lowest mapped values
Soft sage grey	#B7B8A9	Lower middle values
Olive grey	#898A7C	Middle values
Pollution grey	#74746A	Upper middle values
Deep olive grey	#5D5E54	Highest values

Augment with warm white #F7F6F2 for the basemap, teal #247C80 for LSAP dots with a white halo, and charcoal #45483F for ecclesiastical outlines and text. No data uses light neutral hatching, visibly distinct from the lowest class. Use thin boundaries, quiet labels and generous legend spacing. Test contrast, small symbols and greyscale readability before release.

Annual fine particulate air pollution

Map title: Fine particulate air pollution across dioceses. Use an ACAG annual surface PM_{2.5} product, measured in µg/m³. Select the latest complete calendar year shared with the chosen NO₂ data, after verifying availability; show the year in both titles. The two pollutants retain separate legends and units.

Diocesan analytics and story

Report the area-weighted diocesan mean, spatial median and valid land-area coverage. The mean describes the mapped ambient surface across diocesan land, not personal exposure. Optionally report the mean at eligible LSAP member locations as a separate statistic. Do not call it population-weighted without an independent population grid and documented weighting.

Story: How do ambient particulate levels vary across Church territories, and where are participating LSAP communities situated within those patterns? Show the raster at local scale and diocesan summaries at global scale. Explain dust and other natural contributions where relevant rather than attributing every high value to a local industrial source.

Symbology

Use the ordered five-colour ramp, light to dark, with one fixed set of documented breaks across all dioceses for that year. A continuous raster legend should retain numeric units; a choropleth legend should show exact class intervals. Overlay copied diocesan outlines and teal LSAP dots with scale-dependent labels. Avoid opaque polygon fills over the raster.

ModelBuilder style workflow

1. Copy Features → approved dioceses, members and ecclesiastical context into FoundationCopies.gdb; validate keys and copied geometry.
2. Acquire annual ACAG PM_{2.5} → record product version, year, units, native grid, licence, citation and checksum.
3. Prepare Raster → select the annual concentration field, handle fill values as NoData, verify units and land mask; keep native resolution as the analysis basis.
4. Set Environments → snap raster, cell size, extent and mask. Derive equal-area cell or overlap weights; record any reprojection and resampling.
5. Tabulate Intersection or weighted raster aggregation → calculate $\Sigma(\text{concentration} \times \text{valid overlap area}) / \Sigma(\text{valid overlap area})$ for each diocese. Retain coverage and spatial median. A simple unweighted geographic-grid Zonal Mean is insufficient for area weighting.
6. Extract Multi Values to Points → sample the annual raster at eligible copied members; mark NoData and geocoding uncertainty. Summary Statistics → member-location mean and valid member count by diocese.
7. Join Field → append summaries to a fresh diocesan output by stable key; retain every diocese and distinguish null from zero.
8. Apply Symbology From Layer → build raster and diocesan views; check extrema, sample cells, coverage, units and joins. Save final raster, diocesan analytics, copied foundations and workflow metadata.

Annual nitrogen dioxide composite

Map title: Annual atmospheric nitrogen dioxide across dioceses. Use Sentinel-5P TROPOMI offline tropospheric NO₂ column density, in mol/m², for the same complete calendar year as PM_{2.5}. Label it as a tropospheric column, not ground-level NO₂ exposure. The Earth Engine L3 grid is finer than the underlying sensor footprint and does not create finer measurement detail.

Diocesan analytics and story

Summarize the area-weighted annual column mean and valid land-area coverage by diocese. Keep observation counts and seasonal completeness alongside the composite. Story: Where do persistent atmospheric combustion patterns intersect ecclesiastical territories and LSAP communities? Annual averaging reduces short-lived variation but retains cloud, season and sampling limitations.

Symbology

Use the pollution ramp with fixed annual breaks and scientific notation or a clearly labelled multiplier, such as $\times 10^{-5}$ mol/m². Make sparsely sampled areas visibly incomplete. Use identical basemap, boundaries and member symbols to the PM_{2.5} view, while keeping a separate NO₂ legend. Never apply PM_{2.5} health thresholds to the satellite column.

ModelBuilder style workflow

1. Copy Features → reuse verified foundation copies. Select Date → full calendar year, with an exclusive end date on 1 January of the following year.
2. Select Band → tropospheric_NO2_column_number_density from a documented offline product. Record processor versions and the source quality-mask policy.
3. Quality Mask → use the product-recommended QA rules. With Earth Engine L3, record the quality filtering already applied during ingestion; do not claim to reapply a QA band that is absent.
4. Aggregate by Day → combine overlapping valid observations without double-counting split antimeridian assets. Preserve ordinary negative retrieval values unless product guidance identifies invalid outliers; clipping all negatives biases the mean.
5. Aggregate by Month → compute valid daily means and monthly valid-day counts. Annual Composite → combine monthly means weighted by calendar days, retaining coverage; label incomplete composites as observed-period estimates.
6. Coverage Gate → proposed screening: at least 10 valid days in each of at least 9 months, with every calendar quarter represented. Test regional sensitivity and document the final rule; this is a project criterion rather than a universal scientific standard.
7. Weighted Zonal Aggregation → area-weight the composite across valid diocesan land. Calculate valid-area fraction, mean observation count and months represented; flag or suppress inadequate summaries.
8. Extract Multi Values to Points → sample eligible member locations. Join Field → attach diocesan analytics; Apply Symbology From Layer → style separate annual raster and diocesan views.
9. Validate → inspect seams, outliers, temporal coverage and sample values. Save the composite, coverage raster, diocesan tables, foundation copies and processing recipe.

Freshwater quality by diocese

Map title: Freshwater monitoring evidence across dioceses. Use GEMStat water-quality observations with verified reuse and derivative-output permissions. Prefer the open archive where suitable. Source terms vary by submission; removing original records does not independently establish compliance. Retain citations and permission metadata after cleanup.

Diocesan analytics and story

Summarize each parameter separately: for example nitrate, phosphorus, dissolved oxygen or a selected metal. Specify chemical form, units, waterbody type, analytical method and time window. Do not combine unrelated measurements into a universal water-quality score or compare results with unspecified thresholds.

Create a station-year median from comparable valid observations, then a diocesan median of station-year medians. This limits domination by heavily sampled stations. Report distinct station count, represented station-years, sampled years and dispersion where publishable. Proposed release minimum: at least 3 distinct stations and 5 station-years per diocese and parameter; increase it if licence or disclosure conditions require. Small-sample summaries are suppressed, not zero.

Story: What do monitored freshwater sites tell us about each diocese, and where is evidence insufficient? Describe results as monitored-site evidence. Stations do not form a representative census of all diocesan waters. Use parameter-specific map panels and a companion coverage view rather than interpolating a continuous global contamination surface.

Symbology

Fill dioceses with the ordered pollution ramp for one comparable parameter at a time. Where higher values mean better quality, such as oxygen, reverse the concentration-to-pressure interpretation only with a justified threshold framework. Use hatching for insufficient evidence and preserve copied ecclesiastical outlines and approved LSAP dots. Publish no station points or raw-value popups.

ModelBuilder style workflow

1. Rights Gate → record allowed uses, derivative permissions, attribution and deletion requirements before importing observations into restricted FreshwaterScratch.gdb.
2. Select and Standardize → choose parameter, chemical form, waterbody type and stated time window; harmonize compatible units and methods. Handle censored values and detection limits by a documented method, not automatic substitution with zero.
3. Spatial Join → assign stations to copied dioceses using the boundary rule; preserve unassigned and ambiguous counts in restricted QA.
4. Summary Statistics → station-year median for each parameter; then diocesan median, dispersion, distinct station count and represented station-years. Apply the release minimum and coverage flags.
5. Join Field → create aggregate-only diocesan polygons. Keep ecclesiastical and LSAP foundation copies; exclude station IDs, coordinates, individual dates and raw measurements from retained freshwater outputs.
6. Build Map → aggregate choropleths, coverage panel and safe popups. Check that retained summaries cannot expose a single station or reconstruct restricted records.
7. Delete Temporary Data → remove source downloads, extracted records, scratch tables, station layers, joined intermediates and recoverable project copies created for this analysis after aggregate QA. Restrict cleanup to the designated pollution workspace; do not alter unrelated source holdings.
8. Release Audit → inspect project connections, packages, attachments, caches and exports for raw freshwater data. Retain only permitted summarized analytics plus non-record-level provenance, citations, permission evidence and the workflow. Record the cleanup outcome.

River plastic emissions and LSAP members

Map title: River plastic outflows and participating communities. Use Meijer et al. estimated annual floating macroplastic emissions at river outfalls. Preserve the source model scenario, reference period and native units. Confirm field definitions from the downloaded metadata rather than inferring units from field names.

Relationship to the dots

Interpret dots as copied LSAP member locations. Pair outfalls with member dots using nearest geodesic distance and counts within proposed 20 km and 100 km radii. Show emissions at each outfall and the nearest-outfall distance for eligible members. These distances express geographic proximity, not exposure, responsibility, upstream connectivity or member contribution to plastic emissions.

For an optional upstream relationship, link the model outfalls to an independently verified drainage network and delineate their contributing catchments. Spatially assign members to those catchments. Keep this analysis separate from distance buffers; a nearby member may drain to another river. Outfall emissions must not be attributed to the coastal diocese as though all waste originated there.

Symbology

Use deep olive grey proportional circles with pale fill at 40–55% transparency, thin dark rims and graduated symbol areas proportional to emissions. Add a nonzero minimum display size only if disclosed. Use three numeric reference circles in the legend. Place smaller teal LSAP dots above them with white halos. If extreme values require a log-scaled symbol view, label the scale explicitly and retain true values in popups.

Draw selected nearest-outfall connection lines only at local scale, in muted teal; avoid a global web of lines. Popups state “modelled annual outfall emission” and “straight-line geodesic distance”. Use a quiet ecclesiastical outline layer throughout.

ModelBuilder style workflow

1. Copy Features → members and ecclesiastical context. Import Outfalls → preserve original model attributes, identifiers, licence, scenario and reference period.
2. Validate → check point coordinates, emission units, nulls and duplicate outfalls. Retain all outfalls in the analysis, even if low values are hidden in a display-only scale filter.
3. Generate Near Table → geodesic nearest outfall for each eligible member, including across diocesan boundaries. Keep nearest outfall key, distance and modelled emission.
4. Pairwise Buffer or Geodesic Buffer → outfalls at 20 km and 100 km. Spatial Join → count distinct members per outfall and distinct outfalls per member; deduplicate overlapping matches.
5. Summary Statistics → optional diocesan member counts within each radius, valid member denominator and distance distribution. Do not sum overlapping-circle emissions as a physical pollution burden.
6. Optional Catchment Branch → verified outfall-network matching, upstream catchment tracing and member-catchment assignment; label ambiguous matches.
7. Apply Symbology From Layer → proportional emission circles, member dots and ecclesiastical context; create local selected connections and clear distance legends.
8. Validate → sample geodesic distances, member precision, antimeridian behaviour and overlapping buffers. Save source outfalls, proximity derivatives, foundation copies, symbology and workflow metadata.

Landfills and dumpsites by diocese

Map title: Mapped waste sites across dioceses. Use OpenStreetMap landfill footprints, supplemented by documented Waste Atlas dumpsites where reuse permits. Preserve each original pollution layer for symbology and derivatives. Archive the download date, OSM attribution and ODbL obligations; check Waste Atlas permissions before redistribution.

Diocesan count area and member proximity

Report distinct mapped-site count, mapped footprint area in hectares, footprint area as a percentage of diocesan land, and optional sites per 1,000 km². Assign each distinct site once to its primary diocese by greatest footprint overlap, with a reproducible tie-break rule; separately record sites intersecting each diocese. Split footprint area across boundaries and union overlapping footprints to avoid double-counting land.

A point-only dumpsite contributes to site count but has unknown area, never zero area. Deduplicate OSM and Waste Atlas representations using identifiers, location, name and footprint review; retain uncertain matches with flags. Coverage is uneven, so rankings describe mapped inventories rather than actual comparative waste burden.

Measure eligible member distance to the nearest footprint edge, or to the site point when no footprint exists. Count distinct members within proposed 1 km, 5 km and 20 km radii. Distances are geometric proximity, not evidence of contamination or health effects. Keep point-distance results distinguishable from footprint-distance results.

Symbology

Provide toggleable diocesan count, footprint area and area-share views with separate legends. At local scales show retained waste polygons in pollution grey with a deep olive outline; use a small diamond for point-only sites. Differentiate verified controlled landfill, verified open dump and unknown type by outline or shape; never infer type from the generic landfill tag. Place teal member dots above sites and use restrained proximity rings only when selected.

ModelBuilder style workflow

1. Copy Features → foundation layers. Acquire OSM and permitted Waste Atlas records → retain unmodified pollution source snapshots and attribution metadata.
2. Normalize → create source IDs, stable site IDs, geometry type, reported site type, status, date and confidence fields. Repair only analysis copies. Retain closed or disused sites separately from current or unknown-status sites.
3. Deduplicate → cross-source matching and review; Dissolve by Site ID → consolidate multipart sites without merging adjacent distinct facilities.
4. Pairwise Intersect → split polygon sites by diocesan boundaries. Calculate Geometry Attributes → equal-area hectares. Assign primary diocese by greatest overlap; assign point sites by location with explicit boundary rules.
5. Union or Dissolve Coverage → eliminate overlapping footprint area. Summary Statistics → distinct primary-site count, intersecting-site count, known-area count and footprint hectares per diocese.
6. Calculate Field → $100 \times \text{footprint hectares} / \text{diocesan land hectares}$ and optional site density. Join Field → retain all dioceses; label zero as no mapped sites rather than verified absence.
7. Generate Near Table → geodesic member-to-footprint edge distance and a separate member-to-point distance branch. Use 1 km, 5 km and 20 km geodesic buffers → Spatial Join and distinct-member counts; include sites outside a member's diocese.
8. Apply Symbology From Layer → original-site view plus derivative diocesan and proximity views. Validate boundary totals, duplicate handling, missing footprints, member precision and units.
9. Save → retained original pollution layers, cleaned derivatives, aggregate dioceses, member proximity tables, foundation copies and reusable styles.

Map completion and release checks

Each completed map must include the copied ecclesiastical context and approved copied LSAP data, a readable legend, units, time period, source attribution, analytical limitations and the typed workflow updated to the tools and parameters actually used. Keep the maps visually coordinated without implying that their metrics are interchangeable.

Required final outputs

1. Air → annual PM_{2.5} raster and diocesan analytics; annual NO₂ composite, temporal coverage and diocesan analytics; optional member samples, kept as separate metrics.
2. Freshwater → permitted diocesan parameter analytics, coverage and suppression flags only. Retain no original freshwater records, stations or record-bearing intermediates in the final pollution workspace or packages.
3. River plastics → original outfall layer, model attributes, LSAP proximity derivatives and optional independently verified catchment relationships.
4. Waste sites → original OSM and permitted supplementary layers, deduplicated sites, diocesan count and footprint-area derivatives, and member proximity outputs.
5. Foundations → copied member, diocesan and ecclesiastical context layers in each map. Confirm source datasets remained unchanged and output joins preserve every diocesan key.
6. Documentation → per-map source versions, exact input paths, software versions, parameters, classification breaks, QA results, output paths and retention outcome. Workflows describe completed operations only after the map has been executed and checked.

Scientific and cartographic checks

Reconcile counts and area totals, inspect boundary cases, test outliers and NoData, and record valid coverage. Use common breaks within each metric and year, explicit legends for any log transformation, and visible incomplete-evidence states. Test world and local views, antimeridian seams, small dioceses, dense member clusters, print contrast and popup wording. Keep sparse data from becoming a misleading smooth global surface.

Sources and use conditions

ACAG air pollution products and accompanying publications: <https://sites.wustl.edu/acag/datasets/>

Sentinel-5P offline NO₂ product, units and quality processing:

https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S5P_OFFL_L3_NO2

GEMStat archive and coverage: <https://gemstat.org/update-gfqa/> and <https://gemstat.org/data-gemstat/> . Verify the exact submission-level permissions before use. Open and limited-use submissions have different conditions.

Meijer et al. 2021 outfall GIS data: <https://doi.org/10.6084/m9.figshare.14515590> . Use the dataset's citation, licence and model metadata.

OpenStreetMap landfill tagging: <https://wiki.openstreetmap.org/wiki/Tag:landuse%3Dlandfill> . Attribution and licence: <https://www.openstreetmap.org/copyright>

Waste Atlas: <https://www.atlas.d-waste.com/> . Confirm access, reuse and redistribution terms for the specific records retained.